Session Quiz — Train a Small Language Model (BPE + Embeddings)

These multiple-choice questions are for the live Zoom quiz during the session. Each question has one correct answer (marked with **Answer:**) and a short explanation you can use to talk it through.

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1. What is the main job of a tokenizer in an LLM pipeline?

- A) To generate new text
- B) To convert raw text into tokens the model can process
- C) To compute cosine similarity
- D) To train the model's weights

Answer: B — The tokenizer turns human text into the numbered tokens the model actually reads.

2. How does Byte-Pair Encoding (BPE) build its vocabulary?

- A) It assigns exactly one token per character
- B) It starts with characters and repeatedly merges the most frequent adjacent pair
- C) It uses a fixed dictionary of 1000 words
- D) It randomly splits words apart

Answer: B — BPE begins with characters, then merges the most common pairs step by step to form subwords.

3. Why do modern tokenizers use subwords (like "learn" + "ing") instead of whole words?

- A) Subwords make the model slower
- B) They handle rare and unseen words by composing familiar pieces
- C) They remove the need for embeddings
- D) They force the vocabulary to grow into the millions

Answer: B — Breaking words into subwords lets the model understand new words from pieces it already knows.

4. What is a word embedding?

- A) A random list of word IDs
- B) A dense vector that captures a word's meaning as numbers
- C) The raw text of a sentence
- D) A type of attention mask

Answer: B — An embedding is a list of numbers (a vector) where nearby vectors mean similar meanings.

5. Cosine similarity measures the angle between two embedding vectors. A value close to 1 means the two words are:

- A) Unrelated
- B) Opposites
- C) Very similar in meaning
- D) Exactly the same word

Answer: C — A cosine near 1 means the vectors point in nearly the same direction → similar meaning.

6. A cosine similarity close to 0 between two words tells us they are:

- A) Nearly identical
- B) Unrelated / orthogonal in meaning
- C) Translations of each other
- D) Always next to each other in text

Answer: B — A value near 0 means the vectors are at right angles, so the words are not semantically related.

7. In our tiny model, increasing "block_size" (the context window) from 8 to 32 mainly helps the model:

- A) Use less memory
- B) See more of the story before predicting the next token
- C) Remove all punctuation
- D) Train instantly

Answer: B — A bigger context window lets the model look further back before guessing the next word.

8. Increasing "n_embd" (embedding dimension) from 32 to 64 gives each token:

- A) A shorter word ID
- B) A richer vector representation
- C) A fixed position in the sentence
- D) Fewer total parameters

Answer: B — More dimensions = a more detailed vector, so each word can capture more nuance.

9. What does training a small language model actually learn?

- A) It memorizes sentences word-for-word
- B) It learns better embeddings (and weights) to predict the next token
- C) It deletes rare words from memory
- D) It randomly rearranges the tokens

Answer: B — Training adjusts the weights so the model gets better at predicting what comes next.

10. In next-token prediction, what is the model trying to output at each step?

- A) A summary of the whole text
- B) The most likely next token given the context

- C) A completely random word
- D) The very first token again

Answer: B — The model scores every possible next token and picks the most likely one.

11. What does the attention mechanism let the model do?

- A) Ignore the context completely
- B) Weight different context tokens by how relevant they are
- C) Shrink the vocabulary size
- D) Replace the embeddings entirely

Answer: B — Attention lets the model focus more on the tokens that matter for the next prediction.

12. When you scale the same model recipe to more dimensions and more layers, what generally happens?

- A) The parameter count stays exactly the same
- B) The model ends up with fewer parameters
- C) The parameter count and capacity grow (e.g., ~12.9M parameters)
- D) Training becomes instant

Answer: C — More dimensions and layers mean many more parameters, which lets the model capture more patterns.

Answer key (quick reference)

1. B · 2. B · 3. B · 4. B · 5. C · 6. B · 7. B · 8. B · 9. B · 10. B · 11. B · 12. C